Deployment & Setup Guide

ChequePrint Pro — Developer Build & Release

Version: 1.0.0

1. Development Environment Setup

Prerequisites

Tool	Version	Install
Node.js	20 LTS	https://nodejs.org
npm	10.x	Bundled with Node.js
Git	2.40+	https://git-scm.com
Windows Build Tools	Latest	`npm install -g windows-build-tools` (Windows only)

Clone & Install

```
git clone https://github.com/your-org/chequeprintpro.git
cd chequeprintpro
npm install
```

Environment Variables

Create a `.env` file in the project root:

```
NODE_ENV=development
APP_VERSION=1.0.0
```

2. Project Scripts

```
# Start in development mode (Electron + Vite hot reload)
npm run dev

# Run unit tests
npm run test

# Run tests with coverage report
npm run test:coverage
```

```
# Lint code
npm run lint

# Build frontend (Vite)
npm run build

# Build Electron app for current OS
npm run build:electron

# Build installers for all platforms (requires macOS for DMG)
npm run dist:all
```

3. Build Process

3.1 Frontend Build (Vite)

```
npm run build
```

Output directory: `./build/`

3.2 Electron Package

```
npm run build:electron
```

Uses `electron-builder` with config from `electron-builder.config.js`.

Output directory: `./dist/`

Platform	Output File
Windows	`dist/ChequePrintPro-Setup-1.0.0.exe`
macOS	`dist/ChequePrintPro-1.0.0.dmg`
Linux	`dist/ChequePrintPro-1.0.0.AppImage`,`dist/ChequePrintPro-1.0.0.deb`

4. Code Signing

Windows Code Signing

```
# Set environment variables for code signing
export CSC_LINK=./cert/certificate.p12
export CSC_KEY_PASSWORD=your_certificate_password

npm run dist:win
```

macOS Code Signing

```
export APPLE_ID=your@email.com
export APPLE_ID_PASSWORD=your_app_specific_password
export APPLE_TEAM_ID=YOUR_TEAM_ID

npm run dist:mac
```

> **Note:** macOS builds require notarization via Apple's notarization service for distribution outside the App Store.

5. Release Checklist

Before every release:

- [ ] All unit tests passing (`npm run test`)
- [ ] Zero lint errors (`npm run lint`)
- [ ] Version number updated in `package.json`
- [ ] Changelog updated
- [ ] Bank templates verified against latest cheque leaf measurements
- [ ] E2E tests passing on Windows 10, Windows 11
- [ ] Installer size < 120 MB
- [ ] Code signed (Windows .exe, macOS .dmg)
- [ ] SHA-256 hash published alongside download

6. Database Migration

On version upgrades, the database schema may change. Migrations are run automatically on startup via the migration manager:

```
// electron/db/migrate.js
const migrations = [
  {
    version: 1,
    up: `ALTER TABLE cheque_records ADD COLUMN batch_id TEXT;`
  },
  {
    version: 2,
    up: `CREATE INDEX IF NOT EXISTS idx_cheque_date ON cheque_records(cheque_date);`
  }
];
```

Migration version is tracked in `app_settings` table under key `db_version`.

7. Logging

Application logs are written to:

- **Windows:** `%APPDATA%\ChequePrintPro\logs\app.log`
- **macOS:** `~/Library/Logs/ChequePrintPro/app.log`
- **Linux:** `~/.config/ChequePrintPro/logs/app.log`

Log levels: `ERROR`, `WARN`, `INFO`, `DEBUG` (debug disabled in production builds).

Sensitive data (payee names, amounts) is **never logged**.

8. Known Issues & Workarounds

Issue	Affected OS	Workaround
Custom paper size silently ignored on some HP printers	Windows 10	Register custom size via HP driver settings panel manually
AppImage requires FUSE on older Ubuntu	Ubuntu 18.04	`sudo apt install fuse`
SQLite WAL lock on network drives	All	Always store database on local drive, not NAS/network path
macOS 14 (Sonoma) security quarantine	macOS 14	`xattr -d com.apple.quarantine ChequePrintPro.app`